Department of Mechanical Engineering

Faculty of Engineering and Design

FINAL YEAR BEng PROJECT REPORT (ME30227)

HOW DO DIFFERENT FLIGHT STRATEGIES OPTIMISE RANGE OF TIME-IN-THE-AIR FOR A SOLAR POWERED UAV?

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"I certify that I have read and understood the entry in the Student Handbook for the Department of Mechanical Engineering on Cheating and Plagiarism and that all material in this assignment is my own work, except where I have indicated with appropriate references."

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Abstract

Currently, much of the research on Unmanned Aerial Vehicles (UAVs) has focused on high-altitude long-endurance platforms like Zephyr developed by Airbus. Limosaero, a startup, seeks to develop a low-altitude, long-endurance UAV that matches the range and endurance of higher-altitude counterparts while leveraging advantages such as enhanced surface imaging that comes with low-altitude flights. After a preliminary review of existing low-altitude UAVs, the aim of the research were as defined: to investigate factors that affect solar power available to the UAV such as cloud coverage, time of year, and location; to develop a flight model in 2 degrees of freedom to explore how different speed & UAV trajectory modification affects the UAV's performance at summer solstice (Jun-20), equinox (Mar-20) & winter solstice (Dec-21); to determine the optimal flight strategy over a day by varying the speed and trajectory of the UAV.

Python was employed to program the simulations, while images were sourced from the BBC Weather website to fulfil project requirements. Variations in season, time of day, UAV speed, and trajectory were tested, with a focus on evaluating the most efficient flight strategy based on the UAV's battery level upon completing a 65km flight.

Results revealed that flying at the minimum drag speed (12 m/s) proved optimal when no sunlight was available. However, during periods of ample sunlight—particularly during the summer solstice—faster speeds, reaching up to 21 m/s, could be sustained. Notably, speeds of 18 m/s were feasible during the equinox period, and 12 m/s were maintained during the winter solstice. These strategies extended the UAV's range by 313.2 km (1350 km total range) during summer solstice, 118.8 km during equinox (1155.6 km total range) but maintained the same range of 1036.8 km during winter solstice. By modifying trajectory, power savings of 49.61 Wh, 73.01 Wh, and 88.69 Wh could be achieved during summer solstice, equinox, and winter solstice, respectively.

While the project identified an optimal flight strategy, challenges arose in substantiating the benefits of the gained power versus the time lost due to the modified trajectory. Extra time that could have been used to cover further distances. Further studies are required to fully understand and justify if the recommended flight strategy is truly optimal.

Acknowledgements

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Introduction

1.1 Project Background

Research done by business consultancy PwC forecasts that successful adoption of Unmanned Aerial Vehicle (UAV) technology could have 900k + drones operating in the UK skies by 2030 which could contribute 45 billion pounds to the UK economy, introduce 650,000 jobs, and help cut carbon emissions by 2.4 million tonnes. [1]

UAVs have emerged as revolutionary instruments in a variety of industries and applications that were previously limited to traditional manned aircraft. UAVs, with their adaptability, cost-effectiveness, and ability to operate remotely, have evolved from their military origins to become important tools in industries ranging from agricultural and environmental monitoring to disaster response and infrastructure inspection.

Existing low-altitude electric drones have limited range and endurance.

Among the many improvements in UAV technology, the use of solar power stands out as a sign of long-term growth. Solar-powered UAVs combine renewable energy and autonomous flight, providing greater endurance, lower environmental impact, and increased operational flexibility as compared to traditional combustion or jet-powered UAVs. As worries about climate change and environmental sustainability grow, the use of solar-powered UAVs demonstrates a proactive effort to reducing carbon emissions and adopting sustainable energy solutions.

High-altitude solar-powered drones operate above the clouds and are operationally complex.

Currently, extensive research and development efforts have been dedicated to high altitude, long-range solar-powered UAVs, which have led to many successful operations. This technology, exemplified by the collaboration between QinetiQ and Airbus in the Zephyr UAV, operates within the range of 50-70,000 feet and can sustain flight for over 64 days [2][3]. Operating at such heights enables the UAV to evade weather disturbances, ensuring a continuous supply of solar energy. However, it's worth noting that this altitude may not be optimal for all applications, as cloud cover can obstruct optical monitoring of the ground.

Figure 1: Computer generated image of HALE solar UAV Zephyr [2]

Satellites are expensive, their imagery can be obscured by clouds, and they cannot be tasked on-demand.

Low altitude atmospheric monitoring and data collection offer boundless possibilities, driving the increasing demand for Low Altitude Long Endurance (LALE) UAVs. However, LALE UAVs remain relatively uncommon, with current models lacking comprehensive operational testing, especially regarding their performance in varying cloud cover conditions. Current LALE solar UAVs include Scan Eagle capable of flying over 20 hours at 16,000ft [4]. Notable achievements in this field include AC Propulsion's solar-powered SoLong UAV, which flew for 48 hours in California in 2005 [5], and AtlantikSolar's 2015 flight, which flew continuously for 12 hours on battery power, setting a world record for sub-50kg UAVs with an 81-hour flight [6]. Both flights benefited however, from minimal cloud cover and abundant solar irradiance during daylight cycles.

The absence of a commercially available solar-powered LALE UAV that is capable of long endurance flight under any weather conditions underscores a market limitation that Limosaero, the company that this project is based on, aims to address.

1.2 Limosaero

Limosaero are a start-up in Cambridge who aim to develop a sub-20kg solar powered LALE UAV which can be used to detect wildfires, explore rare earths, monitor gas levels, sample the atmosphere, and survey the environments [7]. The UAV will fly below the clouds for several days thus it will have to cope with varying weather conditions such as wind, inconsistent solar irradiance, and fluctuating airmass.

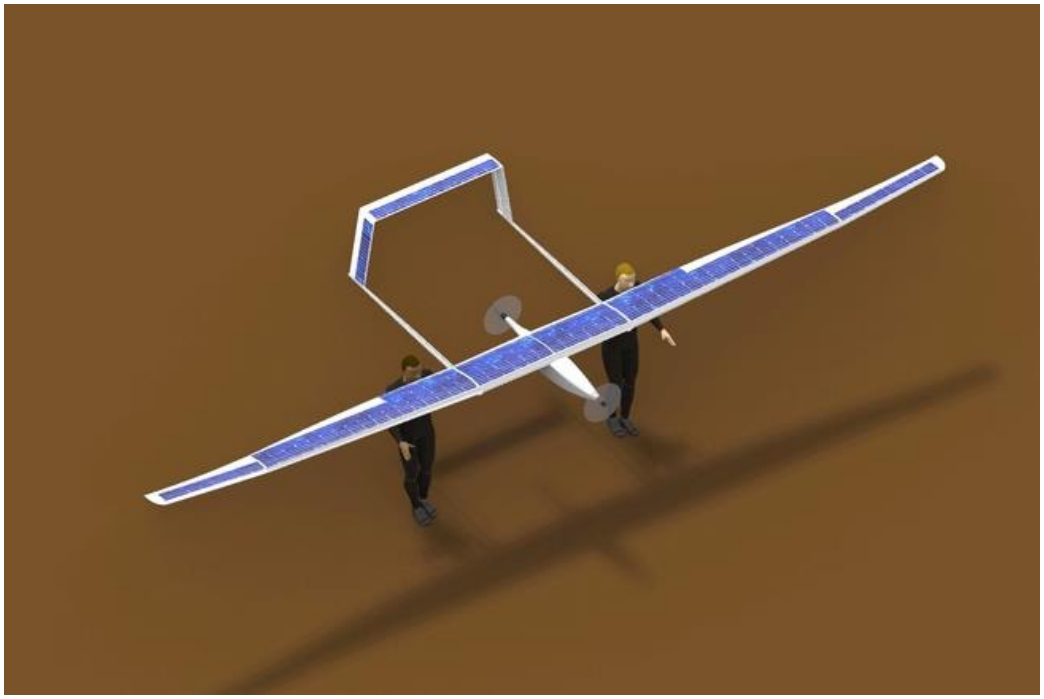


Figure 2: Computer generated image of Limosaero's solar UAV [8]

Limosaero solar UAV has a wingspan of approximately 8m, a mass of 18-20kg and operate at around 40-50 knots. It will be powered by both a pusher and tractor propellers and charge its batteries using its photovoltaic cells which are installed over the entire wingspan and vertical stabilisers. It has a high aspect ratio with a short body and double boom assembly keeping drag minimum improving its performance [8].

However, the ultimate challenge for Limosaero and similar LALE solar UAVs lies in achieving perpetual flight. While flying at minimum drag speed seems like a straightforward solution, the inconsistency of solar irradiance due to cloud cover poses a challenge. Moreover, to maximize the UAV's range, the key is optimizing its performance to capitalize on periods of abundant solar energy effectively during peak solar irradiance like noon times. This demands novel approaches to capturing extra solar irradiance to further improve the range of the UAV.

1.3 Previous Projects

Three previous projects conducted at the University of Bath focused on Limosaero's UAV. The initial project centred on constructing a 3-degree-of-freedom flight model to aid in the development of mission planning algorithms. This model incorporated basic parameters such as solar irradiance, cloud cover, and wind, operating under simplifying assumptions like maintaining constant angle of attack and velocity, as well as fixed propeller power during cruise. [9]

The subsequent project delved into the variability of mission success rates amidst uncertain cloud cover, revealing that introducing a 10% uncertainty factor in cloud cover could lead to fluctuations of up to 5% in mission success. Notably, the percentage of cloud cover held particular significance at latitudes farthest from the equator, leading to the conclusion that forecasts could be utilized for strategic flight decisions. [10]

The third project aimed to enhance the previous model by exploring how knowledge of upcoming cloud cover changes could optimize UAV performance. It was determined that latitude played the most significant role, accounting for up to 60% of performance variation, followed by the time of year at 40%, with the best performance observed during summer and cloud cover contributing up to 35%. [11]

Currently, an ongoing project is investigating the impact of different solar panel arrangements on various UAV types, as well as the orientation of UAV flight concerning incident solar irradiance angle, to discern their effects on flight performance.

1.4 Aims and Objectives

The aim of the project was to provide the best flight strategy by simulating realistic challenges that the company Limosaero might encounter while investigating problems never taken on by previous projects. The project, therefore, aims to take a deeper look at the problems relating to clouds, arguably the greatest challenge to LALE perpetual flight, through introducing trajectory modification for cloud avoidance on top of exploring various speed configuration to determine the optimal strategy to maximise the range of the UAV.

Aims:

- Investigate factors that affect solar power available to the UAV such as cloud coverage, time of year, and location of the UAV.
- Develop a flight model in 2 degrees of freedom to explore how various speed & UAV trajectory modification affects the UAV's performance at various times of the year.
- Determine the optimal flight strategy over a day by varying the speed and trajectory of the UAV.

Objectives:

1. Learn Python programs relevant to the project.
2. Investigate how various factors such as the location & time of year affect the solar irradiance available.
3. Develop a code (code 1) where the user inputs the location, year, month, and date and the code outputs the solar irradiance available under those conditions.
4. Investigate the effects of cloud cover on solar irradiance.
5. Develop a code (code 2) where the user inputs an image of cloud cover in an area and the code outputs a grid of values corresponding to the cloud cover intensity per pixel.
6. Investigate factors that affect the power input and the power output of the UAV.
7. Develop a code (code 3) where a point source moves in a 2 degree of freedom where the inputs are as follows:
    a. Solar irradiance data from code 1 and cloud cover grid data from code 2
    b. The dimension of the area in which the UAV will fly over in meters and the speed of the UAV in m/s
    c. The start and end time of the simulation
    d. The coordinates of the desired trajectory of the UAV

    With the stated input the code runs so that the point source follows the coordinates inputted and at the same time the code calculates the power in (solar irradiance * cloud intensity coefficient at the point source's location * …) and the power out (thrust force * speed) at each tick of the simulation. Finally, the code then outputs:

    a. The total distance travelled.
    b. Final battery level

Literature Review

Since the first solar powered UAV in the 1970s, solar-powered UAV technology has evolved significantly, particularly in the realm of High-Altitude Long Endurance (HALE) solar UAVs. However, academic literature on Low-Altitude Long Endurance (LALE) solar UAVs are yet very limited. Research done by Meyer et al's [12] and Bakar at al's [13], managed to model an aircraft in Simulink which achieved perpetual flight at summer solstice, but the model took liberty of certain factors such as the weather where the model assumed to have 0 cloud cover which wouldn't be realistic under normal conditions. This again highlights the lack of research into LALE solar powered UAVs perpetual flight under realistic weather conditions.

2.1 Solar Irradiance

Assuming that the earth is in perfect orbit around the sun a TSI (total solar irradiance) of 1368W/m^2 is available at the distance of 150.56Km, average distance between the sun and earth. However, TSI is only available when the surface of the earth is perfectly perpendicular to the solar arrays and depending on where the UAV is situated, the 90-degree perpendicular angle may never be achieved. Furthermore, the angle between the surface at a particular location and the solar arrays known as the declination angle changes depending on the season/ time of year. The declination angle reaches a maximum of 23.45° during summer solstice of June 20th in the northern hemisphere and 0° during equinox on March 20th / September 22nd and -23.45° during winter solstice (summer solstice in the southern hemisphere) on December 21st. Furthermore, all the dates mentioned above vary every year, with the above dates representing the relevant celestial dates of 2024. [14]

Thus, to find the peak solar irradiance in each day and location, the latitude is subtracted by the declination angle and the cosine of this new angle is multiplied by the TSI. Given in the northern hemisphere the latitude subtracted by declination angle during summer solstice will yield a value closer to zero degrees thus, peak solar irradiance will be greater during summer and vice versa for either during winter season or in the southern hemisphere. [15]

Next, with the following *equations 1-5*, the solar noon, sunrise, and sunset time can be calculated. (δ: declination angle, $\emptyset$: latitude, EoT: Equation of Time). This gives all the necessary components to find the solar irradiance in a given day at any time of the year.

$Equation\ 1$: $$EoT = 9.81 \sin 2\delta - 7.53 \sin \delta - 1.5 \sin \delta$$

$Equation\ 2$: $$Solar\ Noon = \frac{(720 - EoT)}{1440}$$

$Equation\ 3$: $$H_0 = \cos^{-1}(-\tan \emptyset \times \tan \delta)$$

$Equation\ 4$: $$Sunrise\ Time = Solar\ Noon - \left(\frac{H_0}{360}\right)$$

$Equation\ 5$: $$Sunset\ Time = Solar\ Noon + \left(\frac{H_0}{360}\right)$$ [16]

(All the equations have been slightly altered to simplify some factors such as longitude and to output the final noon, sunrise, and sunset times in minutes rather than seconds or hours. This decision will later be explained in the report.)

2.2 Cloud Properties

Cloud elevation varies depending on latitude with it ranging from 6,500 feet to 60,0000 feet in tropical regions (0° to ±23.5° latitude), 6,500 feet to 45,000 feet in temperate region (±23.5° to ±66.5° latitude) to in between 6,500 feet to 25,000 feet in polar region (>±66.5° latitude). For the sake of simplicity all simulations in the project will assume that the UAV is flying 2000 meters (~=6500 feet) above sea level hence regardless of location the UAV will always experience the full effect of cloud cover. [17]

Table 1: Cloud Altitude Range at various Latitudes [17]

Level	Polar Region	Temperate Region	Tropical Region
High Clouds	10,000-25,000 feet (3-8 km)	16,500-45,000 Feet (5-14 km)	20,000-60,000 feet (6-18 km)
Mid Clouds	6,500-13,000 feet (2-4 km)	6,500-23,000 feet (2-7 km)	6,500-25,000 feet (2-8 km)
Low Clouds	Surface-6,500 feet (0-2 km)	Surface-6,500 feet (0-2 km)	Surface-6,500 feet (0-2 km)

Next, clouds on average move in between 12 to 17 m/s. This is problematic since the clouds will be moving at a similar rate compared to the UAV thus, the direction in which the UAV travels in comparison to the cloud would have a great impact. Furthermore, because the UAV will be flying right under the clouds the relative speed will be the same. For the sake of simplicity again, the project will assume the clouds are still during simulations. [18]

Lastly, clouds can reflect up to 70% of incoming solar radiation, thus for the project it will be assumed that at maximum cloud intensity, only 30% of the solar irradiance will reach the UAV below the clouds. [19][20]

2.3 UAV Power Output

Lastly, to determine the power required for the UAV to fly at certain speeds requires the drag force experienced by the UAV at that speed. The total drag force is a combination of induced drag force and friction/parasite drag force.

$Equation\ 6$: $$Friction\ Drag\ Force = \frac{1}{2}\rho V^2 C_D S$$ [21]

$Equation\ 7$: $$Induced\ Drag\ Force = \frac{W^2}{\frac{1}{2}\rho V^2 \pi b^2 \varepsilon}$$ [22]

$Equation\ 8$: $$Power = (Friction\ Drag\ Force + Induced\ Drag\ Force) \times Velocity$$

With the *equation 6 & 7*, thus the total power consumed per second in watts is given above. Understanding the two equations will help determine the minimum total drag force will be given at the minimum drag speed. Determining the minimum drag speed will be crucial in determining the optimal flight strategy.

All the constants necessary to determine the total power required to overcome any drag force is given in the table below. To determine the density at various altitudes the density-altitude equation can be used derived from the barometric formula. Lift

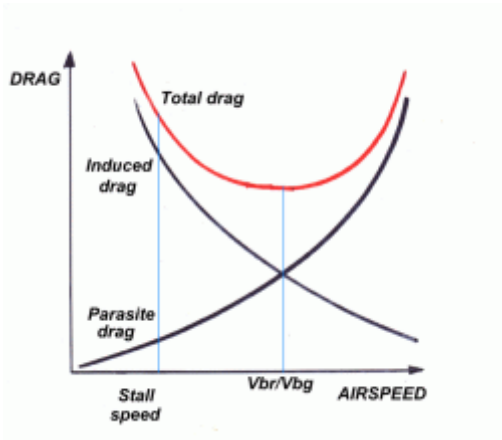


Figure 3: Various Drag Forces against Velocity [22]

and drag coefficients will be determined to be constants and the optimum lift coefficient and the drag polar equations for the Limosaero UAV are derived from the previous report on the UAV [11]. Lastly, the Oswald factor was an unknown quantity thus was assumed to be 0.85 which was common value for a fixed-wing aircraft. [23]

Table 2: Constants needed to calculate Drag Force [11]

Symbol	Parameter	Value
ρ	Density	"Density-altitude Equation"
V	Velocity	Speed Inputted
C_D	Drag Polar	$C_D = 0.00661 + 0.0287{C_L}^2$
C_L	Lift Coefficient	0.4797
S	Solar Panel Area	$2.875m^2$
W	Weight	$20 \times 9.871 = 196.2$
b	Wingspan	8m
ε	Oswald Factor	0.85
$n.a.$	Battery Capacity	1275Wh
$n.a.$	Solar Panel Efficiency	20%
$n.a.$	Propeller Efficiency	72%
$n.a.$	Other Power Draw	10W

2.4 Significance of Research

Currently, research regarding trajectories and their impact on UAV performance is lacking, with most projects simply aiming to move from point A to B in a straight line. Additionally, the effects of cloud cover are often oversimplified, with average cloud cover percentages used to estimate solar irradiance for entire days. However, cloud cover is rarely constant and varies throughout the day, leading to unrealistic experimental conditions.

The project seeks to address these challenges by examining the effects of cloud cover on a micro scale. By analysing short and long-term flight strategies, the project aims to accurately identify problems and propose realistic solutions. In the future, the company envisions having constant surveillance and control over the UAV, enabling real-time adjustments to its trajectory, as well as implementing a general speed strategy for a more hands-off approach. This comprehensive approach aims to optimize UAV performance under varying atmospheric conditions, ensuring more efficient and reliable operations.

Methods

3.1 Solar Irradiance

The code has been developed to calculate key solar parameters essential for optimizing solar-powered systems. The program takes user inputs such as latitude, date, month, and year and establishes constants like Total Solar Irradiance and earth's tilt angle. Then, based on the input date and year, the code determines whether it's a leap year and calculates the corresponding day of the year giving a number from d=1 (Jan 1st) to d=365 (dec 31st). Using this information, the solar declination angle is computed using the equation below.

Equation 9: $$\delta = 23.45 \sin(\frac{360}{365}(d - 81))$$

With the solar declination angle determined, the code proceeds to calculate solar irradiance at the specified latitude by computing the Equation of Time (EoT), solar noon, sunrise, and sunset times with the afore mentioned *equation 1 - 5*. With all necessary information solar irradiance is modelled using the sin function where peak solar irradiance is TSI multiplied by cos of declination angle which happens at noon, and the period of the sin function is twice the value of the difference between sunset and sunrise time.

It was decided to have a total of 1440 data points over the half sine function cycle to represent the 1440 minutes that exists in a 24-hour day. This was a deliberate decision as utilizing seconds would require 86,400 data points which would become too much data for the later codes and would be hard for the system to deal with when multiple data lists become involved.

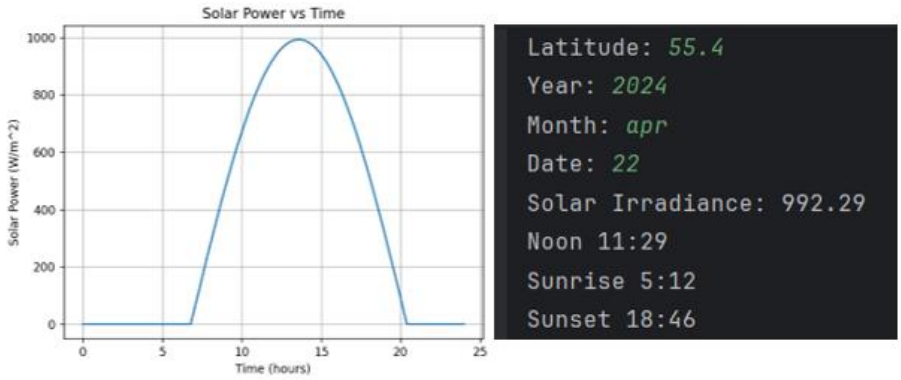


Figure 4: Solar Irradiance Code input & output

3.2 Cloud Imaging

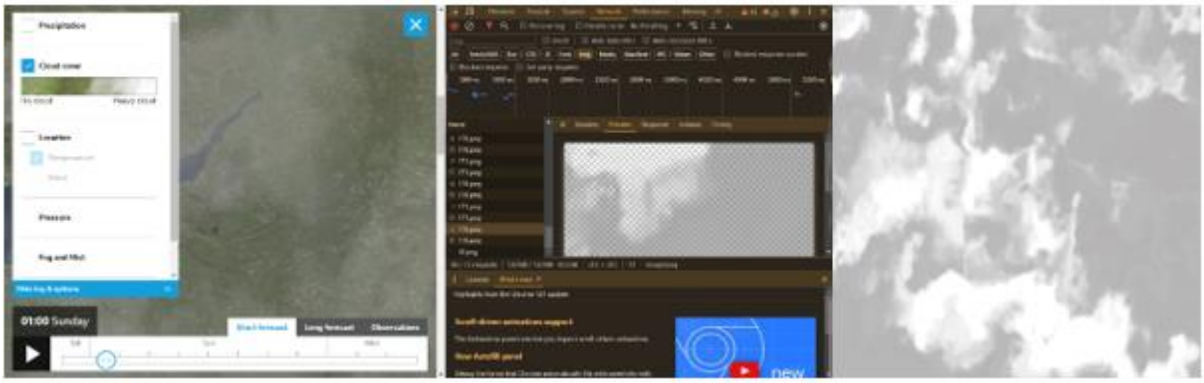


Figure 5: Process of attaining cloud coverage image from weather websites [24]

The above sample image of cloud imagery taken by satellites can be found on most weather sites such as the BBC weather [24]. By using the inspect feature then going to network and images the user can download various cloud image samples over a specific area. Next, to identify the dimensions of the area that each cloud image is supposed to represent some estimations were used. Because it was known that the minimum distance between two images was represented by the area north Brecon and south of Tiverton the rough distance was found using Google Maps. Given the estimated distance was 130km the dimensions of each image were estimated to be 65km x 65km. (Although this is a poor method of approximating the distance of the image give the fact that the project is more focused on finding the effect of cloud coverage and less importance on the precise proportions of the cloud this rough estimation was deemed suitable for the method.) All the cloud image examples in this project were attained from BBC weather where all the images provided were 512 by 512 pixel.

3.3 Cloud Change

On each of the cloud image clouds are represented by a spectrum of the colour grey with thinner clouds being darker grey and regions with no cloud represented as transparent pixels. To produce a code that scanned the 512 by 512-pixel image and create a grid of 512 by 512 values with each value corresponding to the intensity of the cloud, first the code imported cv2 for image processing and NumPy for numerical operations. The first function is coded to

convert any transparent pixels into black pixels to avoid confusions in later calculations involved. This was done by first taking the image with an alpha channel (RGBA) and converts the transparent pixels to black by setting the values of the RGB channels to 0 where the alpha channel is 0. Then the dimensions of the image are obtained, and the image is converted to greyscale. The cloud coverage values are computed based on specific colour thresholds since the lightest colour in the image isn't a perfect white but light grey. The lightest pixel was scanned giving the colour value r=142, g=142, b=142. To get the 512 by 512 grid of cloud intensity coefficient which will be used for later codes, the cloud pixels were scanned and scaled to give a value of 1 (100% of the solar irradiance transmittance) for no clouds/ black pixels and 0.30 (theoretical maximum solar threshold for overcast) for colours with r=142, g=142 b=142. Any pixels with colours in between would be arranged a value in between 1 and 0.3 accordingly giving a spectrum of grid values.

```
Grid of Cloud Coverage Values (Scaled):
[[0.62584507 0.62584507 0.62584507 ... 0.65985915 0.65985915 0.65985915]
 [0.62584507 0.62584507 0.62584507 ... 0.65985915 0.65985915 0.65985915]
 [0.62584507 0.65985915 0.62584507 ... 0.65985915 0.65985915 0.69873239]
 ...
 [0.65985915 0.65985915 0.65985915 ... 0.85422535 0.85422535 0.85422535]
 [0.65985915 0.65985915 0.69873239 ... 0.93683099 0.93683099 0.93683099]
 [0.65985915 0.65985915 0.65985915 ... 0.93683099 0.93683099 0.93683099]]

Grid of Cloud Coverage Values (Normalized):
[[0.54225352 0.54225352 0.54225352 ... 0.49295775 0.49295775 0.49295775]
 [0.54225352 0.54225352 0.54225352 ... 0.49295775 0.49295775 0.49295775]
 [0.54225352 0.49295775 0.54225352 ... 0.49295775 0.49295775 0.43661972]
 ...
 [0.49295775 0.49295775 0.49295775 ... 0.21126761 0.21126761 0.21126761]
 [0.49295775 0.49295775 0.43661972 ... 0.0915493  0.0915493  0.0915493 ]
 [0.49295775 0.49295775 0.49295775 ... 0.0915493  0.0915493  0.0915493 ]]

Overall Cloud Coverage Percentage (Normalized): 0.4046865651305292

Area Dimension (Width x Height): (512, 512)
```

Figure 6: Grid output of cloud intensity coefficients

3.4 Flight Power

With all the necessary data and equations, the battery level over time can be found. Upon specifying the start time of the simulation in between 0 to 1440 the corresponding solar irradiance value at the specified time can be imported. As the simulation keeps running the solar irradiance value would be updated based on time/ game tick. Furthermore, the absolute value of the coordinates of the UAV is taken and inputted into the grid of cloud coverage values to get the corresponding cloud intensity coefficient at that pixel/ coordinate. By multiplying the solar irradiance at that moment of simulation by the cloud intensity coefficient then applying the solar panel efficiency then multiplying by the Limosaero's solar panel area then dividing by 60, the power input of the UAV per minute can be calculated.

```
for i in range(start, end):
```

```
# Calculate powerIn based on the cloud coverage at the square's coordinates
if 0 <= abs(square_x) < 512 and 0 <= abs(square_y) < 512:
    cloud_value = cloud_coverage_values_scaled[int(abs(square_y))][int(abs(square_x))]
    powerIn[i - start] = cloud_value * solarPower[i] * sEff * area / 60
```

Figure 7: UAV power input factoring solar irradiance & cloud intensity coefficient

For the power output of the UAV, it is much simpler since speed of the UAV is constant through a simulation, the induced and friction drag forces can be calculated with ease by utilizing the constants listed in *table 2.*

```
# Calculate drag force
fDrag = 0.5 * density * dCoef * area * speed ** 2

iDrag = (mass * 9.81) ** 2 / (0.5 * density * speed ** 2 * math.pi * span ** 2 * oswald)

# Calculate power consumption
powerOut[i - start] = ((fDrag + iDrag) / pEff) * speed / 60

# Update battery status
battery_level = battery[-1] + powerIn[i - start] - powerOut[i - start] - (passivePower / 60)
```

Figure 8: UAV power output & battery level

With a list of power in and power out values in watt hours, the battery life is calculated by taking the initial battery value of 1275Wh then adding power in and subtracting the power out and passive power draw. The list of all three values over time is plotted and shown in *figure 9* below.

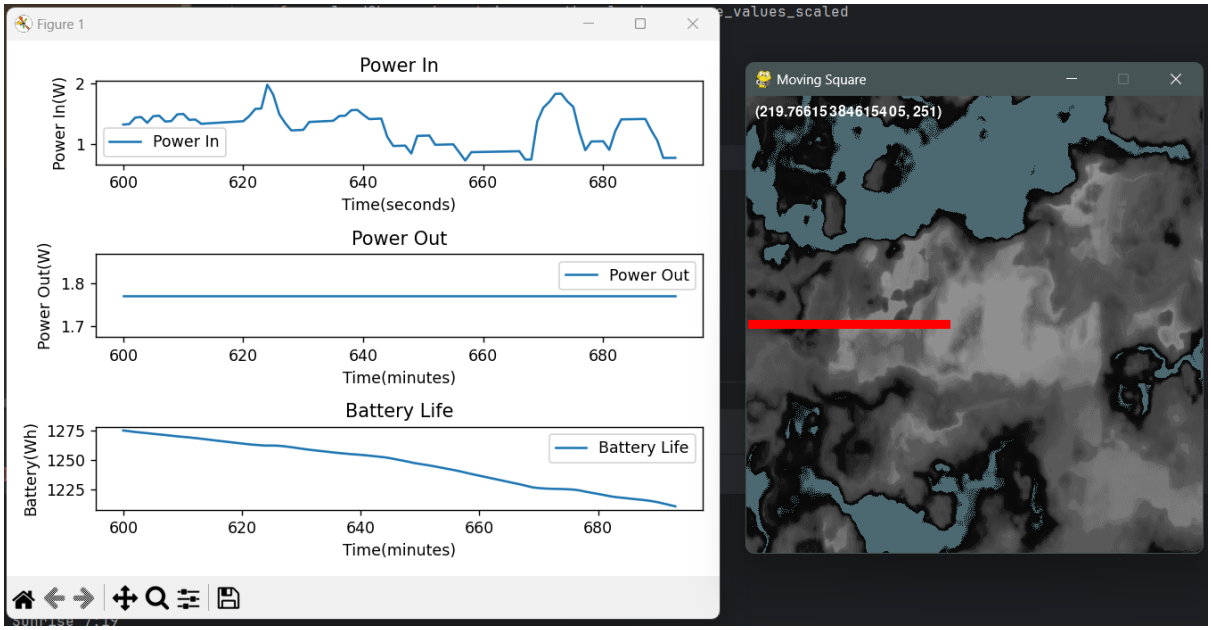


Figure 9: Example of a typical simulation with live plot of input power, output power and battery level

3.5 Flight Simulator

The code utilizes pygame from the library to model a movement of a square within a defined area, giving a 2 degree of freedom model. The code begins by importing necessary libraries including pygame, sys, and modules from cloudChange. The user inputs height and width of the area, speed of the moving dot, and the start and end times. When the code is run, pygame is initialized and a window is set up with a background image of the relevant cloud cover image. Constants and variables related to the square's movement such as its initial position, velocities are initialized and lists are set up to store data related to solar power, power input, power output, and battery level. Constants from the above are defined. Because each game tick represents a minute, velocity is first converted to meters per minute then converted again to pixels per minute with the inputted dimensions of the area and the knowledge that the image represented is 512 by 512 pixels.

The main game loop iterates over time intervals from start to end times. And the user inputs the desired trajectory by inputting the coordinates and the desired direction for the UAV at that coordinate. The code is programmed so that if the user wants the code the UAV to move north at x = 100, the code executes it by checking if the UAV's x coordinate is greater than 100, thus in actuality the UAV always overshoots x=100 and turns north couple meters too late. The square's position is updated based on its velocity in respect to the inputted area dimension and the code constantly checks that the square remains within the window boundaries. A trail is added to the square for aid in trajectory visualization. Cloud coverage at the square's position is obtained to calculate power input based on solar power with cloud coverage varying depending on the position of the UAV, and solar power varying based on time.

```
# Check conditions for changing direction
if square_x >= 10 and not x_100_flag:
    x_velocity = 0
    y_velocity = -speed_y
    x_100_flag = True
elif square_y <= 125 and not y_100_flag:
    x_velocity = speed_x
    y_velocity = 0
    y_100_flag = True
elif square_x >= 400 and not x_200_flag:
    x_velocity = 0
    y_velocity = speed_y
    x_200_flag = True
elif square_y >= 252:
    x_velocity = speed_x
    y_velocity = 0
```

Figure 10: Trajectory Modifier

After the simulation loop ends either because the loop time has reached the end, or the square has reached the window boundary or because battery has dropped to 0, the program displays the distance travelled which is a simple speed * time elapsed in the loop and the final battery level of the UAV.

Results and Analysis

Planning how the results would be taken, it became apparent that time would be the greatest limit due to the sheer number of simulations that would have to be ran to get a satisfactory analysis. Therefore, although the code permitted countless possibilities and conditions for the UAV to be tested under, two major decisions were made. First, the project would only focus on latitude 44.5° (UK's average latitude) and the UAV would be tested under 6 different speed configurations only (5, 10, 15, 20, 25, 30 m/s).

Because the simulation was only interested in going from point A to point B which spanned 65km in the simulations, all flights were expected to successfully complete the mission. Therefore, rather than utilizing parameters such as time taken from point A to B to measure the best flight strategy, the final battery level at when the UAV reaches point B was determined as the best measure to determine the optimum flight strategy.

4.1 Effects of solar irradiance and speed on battery level.

In the initial section of the analysis, the project aimed to observe the overall impact of different speeds throughout the day on the UAV's battery level. Simulations were conducted under the same cloud image, focusing solely on variations in speed and time of day. The UAV followed a direct path from point A to B. Notably, the chosen cloud cover image for this portion of the simulation featured the highest density of clouds, providing a foundation to demonstrate the negative effects of cloud cover. This approach allows for a clearer understanding of the benefits of cloud avoidance strategies in subsequent analyses.

Table 3: Final battery level at various speed and hours for a straight trajectory on summer solstice

Jun-20		Speed (meters per second)					
		5	10	15	20	25	30
Time (hour)	0	864.16	1119.47	1118.73	1055.78	963.15	840.93
	3	1061.59	1129.8	1118.73	1055.78	963.15	840.93
	6	1275	1275	1275	1184.7	1061.53	919.73
	9	1275	1275	1275	1275	1167.27	1007.86
	12	1275	1275	1275	1275	1200.09	1036.35
	15	1274.65	1275	1275	1275	1148.29	995.05
	18	961.12	1217.07	1211.63	1134.47	1030.35	898.7
	21	864.16	1119.47	1118.73	1055.78	963.15	840.93

Here we see a table of results at latitude 44.5 on June 20 the summer solstice where sunlight is the most abundant in the northern hemisphere. Each data represented in the *table 3* represents the final battery value after running the simulations under the stated conditions.

All the simulations that commenced at 12am and 9pm gave the lowest possible battery level possible after 65km of flight. These values vary depending on the speed with the greatest performance at 10 m/s as the performance decreased with increasing speed and also with decreasing speed. The decreasing performance with speed can be explained by the increasing friction drag force, requiring greater power to overcome. While performance decreases with decreased speed due to increased induced drag force which is a consequence of lift produced by passage of aerofoil therefore with decreased air passage drag is increased. The optimum speed to minimize drag can be described as the point in which both frictions drag force and induced drag force is the minimum and is known as the minimum drag speed. In the case of the Limosaero UAV, the minimum drag force is in between 10 to 15 m/s. Therefore, under no sunlight conditions it is optimum to fly at minimum drag speed in a straight line because there are no incentives to avoid clouds. All the conditions in which such flight strategy is optimum will be marked with orange on the table through the entire analysis.

Flying at minimum drag speed isn't the most optimal strategy however as we can see at certain times of the day the UAV is receiving more energy than it is utilizing thus the battery level remains at maximum capacity even under the thickest clouds chosen for this simulation. Under these conditions it is again the most optimal to fly in a straight path since avoiding clouds wouldn't provide any benefits since the battery remains full. It is however most optimal to fly faster than the minimum drag speed as visible in the table marked by yellow regions in which solar power is abundant enough to achieve perpetual flight.

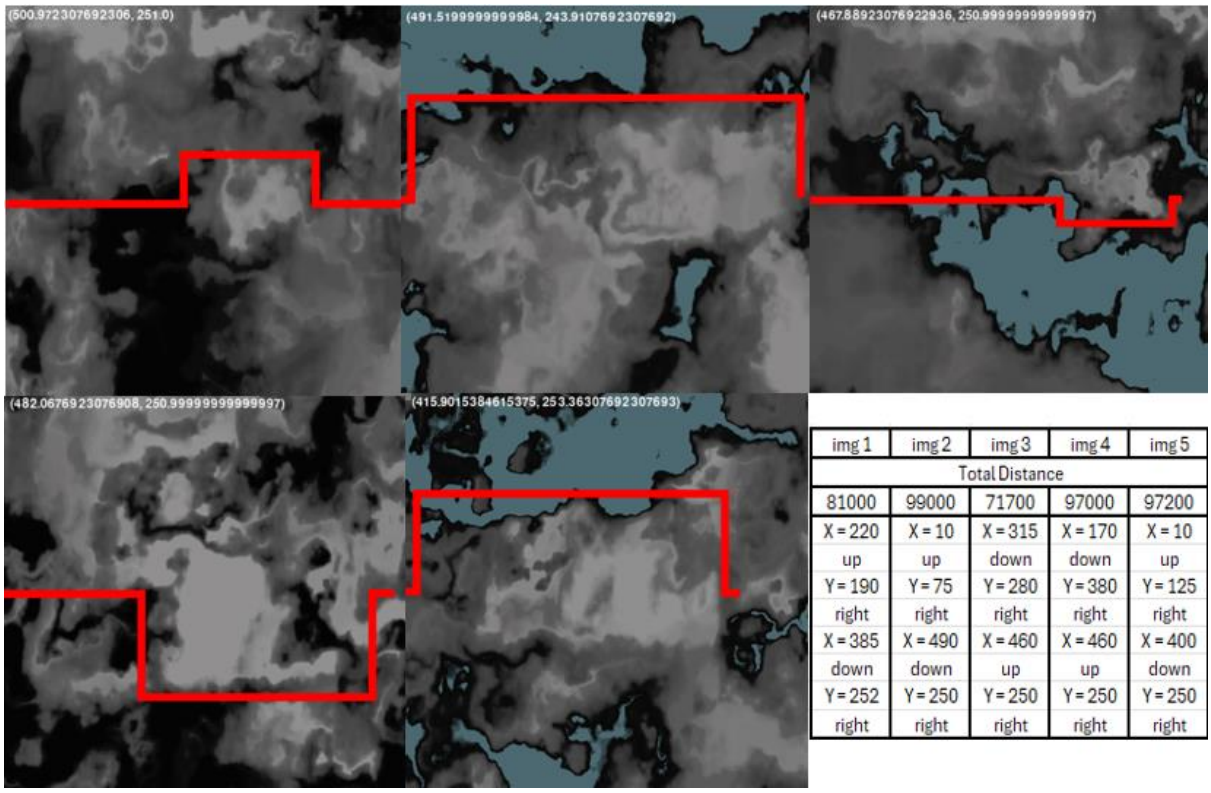
Table 4: Final battery level at various speed and hours for a straight trajectory on equinox & winter solstice

	Speed (meters per second)						
Time (hour)	Mar-20	5	10	15	20	25	30
	0	864.16	1119.47	1118.73	1055.78	963.15	840.93
	3	868.41	1119.47	1118.73	1055.78	963.15	840.93
	6	1193.02	1196.9	1148.83	1071.02	971.47	845.85
	9	1275	1275	1275	1204.51	1078.97	934.96
	12	1275	1275	1275	1250.4	1118.49	969.08
	15	1094.67	1275	1272.03	1180.4	1065.49	926.88
	18	864.41	1119.7	1118.96	1056	963.36	841.14
	21	864.16	1119.47	1118.73	1055.78	963.15	840.93
	Speed (meters per second)						
Time (hour)	Dec-21	5	10	15	20	25	30
	0	864.16	1119.47	1118.73	1055.78	963.15	840.93
	3	864.16	1119.47	1118.73	1055.78	963.15	840.93
	6	915.75	1119.48	1118.73	1055.78	963.15	840.93
	9	1097.71	1217.2	1176.31	1096.31	993.86	865.39
	12	1056.81	1245.15	1205.95	1122.42	1016.6	885.15
	15	882.93	1138.04	1138.29	1074.18	979.7	855.57
	18	864.16	1119.47	1118.73	1055.78	963.15	840.93
	21	864.16	1119.47	1118.73	1055.78	963.15	840.93

Above are the tables of result at the same latitude but on Mar 20 equinox and the winter solstice on Dec 21 represented by the second table. With peak solar irradiance being 1160.66 W/m^2 on summer solstice, 760.91 W/m^2 on equinox and 264.56 W/m^2 according to code 1 and it isn't surprising to see a noticeable change in the performance of the UAV with shrinking yellow zones (conditions in which perpetual flight is possible) to non-existent during winter solstice which also sees increases in orange zones. It is the results in between the orange and yellow conditions that the project is most interested in because those are the conditions in which change of trajectory can affect the performance of the UAV.

4.2 Effects of clouds at constant solar irradiance

A UAV consumes power to travel a greater distance therefore any extra distance the UAV takes to avoid clouds would consume greater power. To know for sure if adjusting the UAV trajectory would lead to enhanced performance, this time the simulations were ran twice, first, in a straight line and second, in a modified trajectory set by the user. Finally, the simulations were under constant solar irradiance condition ranging from 100, 300, 500, 700, 900, 1100W/m^2 where each condition were tested. To first achieve the simulations five images were chosen from a large cloud cover sample based on the size of the clouds and the location of it. The chosen images all had some form of cloud in the UAV flight path and could be navigated around. Flight trajectory that was deemed the most suitable to avoid as many clouds was chosen then the coordinates and the direction instructions were encoded into the simulation. Here are the five instructions and what the trajectory looked like.



img 1	img 2	img 3	img 4	img 5
Total Distance				
81000	99000	71700	97000	97200
X = 220	X = 10	X = 315	X = 170	X = 10
up	up	down	down	up
Y = 190	Y = 75	Y = 280	Y = 380	Y = 125
right	right	right	right	right
X = 385	X = 490	X = 460	X = 460	X = 400
down	down	up	up	down
Y = 252	Y = 250	Y = 250	Y = 250	Y = 250
right	right	right	right	right

Figure 11: Modified trajectories of the five cloud cover plane and their exact coordinates

The sets of simulations were run five times for each of the images above.

Table 5: Comparison of final battery level under different trajectories and various solar irradiance constants

100W/m^2	5	10	15	20	25	30
straight	989.008	1181.85	1159.97	1087.06	987.906	861.534
modified	922.85	1168.26	1129.15	1023.8	878.358	692.43
300W/m^2	5	10	15	20	25	30
straight	1207.53	1269.8	1245.93	1149.63	1037.42	902.752
modified	1254.9	1274.8	1259.78	1133.57	967.172	767.494
500W/m^2	5	10	15	20	25	30
straight	1275	1275	1275	1210.01	1086.94	970.714
modified	1275	1275	1275	1237.61	1055.98	842.558
700W/m^2	5	10	15	20	25	30
straight	1275	1275	1275	1262.52	1136.45	985.184
modified	1275	1275	1275	1273.56	1144.79	917.622
900W/m^2	5	10	15	20	25	30
straight	1275	1275	1275	1275	1183.37	1026.4
modified	1275	1275	1275	1275	1222.36	992.69
1100W/m^2	5	10	15	20	25	30
straight	1275	1275	1275	1275	1216.64	1067.62
modified	1275	1275	1275	1275	1256.43	1067.75

From the table it can be concluded that certain solar irradiance levels are required for the extra distance covered under greater exposure to solar irradiance for the new UAV trajectory to improve the UAV performance. These simulations are marked with green and simulations in which the modified trajectory led to worse performance are marked red. What was interesting was that the greater the distance the UAV had to cover for the new trajectory lead to a greater improvement in final battery level but at the same time under the wrong conditions lead to a greater penalty in performance as seen in *figure 12 & 13.*

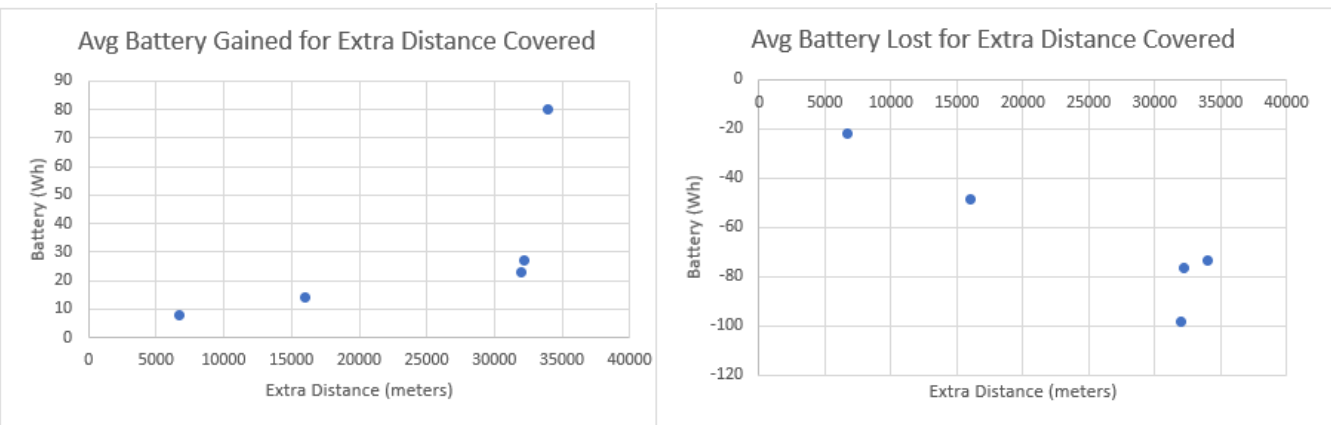


Figure 12: Average battery life gained and lost in watt hour per extra distance covered for the modified trajectories

The table above divided the simulations on whether it improved the battery life (green simulations) or whether it reduced it (red simulations) and took the average value of the battery gained / lost per each image simulation case.

The trajectory with additional distance is likely to encounter larger clouds along the original straight path. Therefore, avoiding these clouds would result in enhanced performance. However, if the conditions are unfavourable—where the power needed to sustain the UAV's speed exceeds the available solar power—flying the extra distance could lead to battery power wastage. This would result in insufficient power to make the extended trajectory beneficial overall.

4.3 Simulation under normal conditions

Next step was now to test the enhanced trajectory under normal day cycle conditions where solar irradiance varies through time. This time the simulations would be ran on a per hour basis again ran in pairs with straight & modified trajectories.

The *tables 3&4* were expanded to get a better understanding of the boundaries of the yellow and orange simulation conditions but also simplified based on the accumulated knowledge from the previous simulations. First example was to remove 5 m/s speed for all simulations since going faster requires less power thus will cover greater range while utilizing less power. Secondly, 30 m/s was deemed too fast and inefficient for it to be considered optimal strategy especially since 1100W/m^2 of solar irradiance is almost impossible to come by at 44.5 latitude. Under a similar philosophy peak speed has been reduced to 25 m/s for summer solstice, 20 m/s for equinox and 15 m/s for winter solstice. Now all that remained was to test all the conditions in between the yellow and orange zones for the 5 different cloud conditions with 2 pairs of simulations. One with a straight path and one with the modified trajectory for comparison.

Table 6: Final Battery Level at various speed and time of day with and without modified trajectory on summer solstice

	Speed (meters per second)								
	Jun-20	10		15		20		25	
Time	0	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78	963.15	963.15
	1	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78	963.15	963.15
	2	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78	963.15	963.15
	3	1127.43	1127.25	1118.73	1118.73	1055.78	1055.78	963.15	963.15
	4	1206.13	1270.25	1149.94	1149.55	1070.41	1007.65	963.15	963.15
	5	1275	1275	1239.83	1275	1139.81	1136.34	1024.76	951.146
	6	1275	1275	1275	1275	1208.94	1253.82	1079.85	1051.15
	7	1275	1275	1275	1275	1251.22	1274.27	1130.18	1141.8
	8	1275	1275	1275	1275	1275	1275	1173.7	1215.88
	9	1275	1275	1275	1275	1275	1275	1200.32	1245.86
	10	1275	1275	1275	1275	1275	1275	1216.33	1258.51
	11	1275	1275	1275	1275	1275	1275	1224.58	1260.31
	12	1275	1275	1275	1275	1275	1275	1225.44	1259.11
	13	1275	1275	1275	1275	1275	1275	1218.98	1253.71
	14	1275	1275	1275	1275	1275	1275	1205.29	1238.2
	15	1275	1275	1275	1275	1275	1275	1184.21	1208.68
	16	1275	1275	1275	1275	1249.74	1258.58	1148.1	1148.72
	17	1275	1275	1275	1275	1214.45	1219.22	1100.21	1058.99
	18	1222.2	1175.59	1226.16	1202.21	1155.21	1114.47	1046.72	959.614
	19	1148.5	1085.38	1147.26	1084.22	1083.46	992.816	989.822	857.552
	20	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78	963.15	963.15
	21	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78	963.15	963.15
	22	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78	963.15	963.15
	23	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78	963.15	963.15

Table 7: Figure 13: Final Battery Level at various speed and time of day with and without modified trajectory on equinox

	Speed (meters per second)						
	Mar-20	10		15		20	
Time	0	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78
	1	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78
	2	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78
	3	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78
	4	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78
	5	1130.33	1123.12	1118.73	1118.73	1055.78	1055.78
	6	1200.29	1262.7	1149.72	1143.44	1068.64	1006.62
	7	1275	1275	1228.38	1271.23	1131.19	1116.12
	8	1275	1275	1270.67	1275	1187.36	1216.23
	9	1275	1275	1275	1275	1227.25	1264.18
	10	1275	1275	1275	1275	1248.17	1270.61
	11	1275	1275	1275	1275	1256.24	1271.96
	12	1275	1275	1275	1275	1256.86	1270.35
	13	1275	1275	1275	1275	1250.13	1264.56
	14	1275	1275	1275	1275	1234.29	1249.49
	15	1275	1275	1265.59	1261.82	1206.46	1213.5
	16	1207.97	1193.83	1230.95	1216.53	1157.31	1122.95
	17	1166.01	1107.89	1163.23	1106.91	1097.75	1014.61
	18	1119.81	1052.62	1119.07	1049.87	1056.11	957.672
	19	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78
	20	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78
	21	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78
	22	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78
	23	1119.47	1119.47	1118.73	1118.73	1055.78	1055.78

Table 8: Final Battery Level at various speed and time of day with and without modified trajectory on winter solstice

	Speed (meters per second)				
	Dec-21	10		15	
Time	0	1119.47	1119.47	1118.73	1118.73
	1	1119.47	1119.47	1118.73	1118.73
	2	1119.47	1119.47	1118.73	1118.73
	3	1119.47	1119.47	1118.73	1118.73
	4	1119.47	1119.47	1118.73	1118.73
	5	1119.47	1119.47	1118.73	1118.73
	6	1119.48	1065.11	1118.73	1118.73
	7	1135.69	1114.04	1121.61	1066.93
	8	1183.38	1201.71	1150.72	1125.46
	9	1232.74	1268.52	1186.03	1186.35
	10	1254.99	1271.44	1212.17	1228.63
	11	1259.78	1267.03	1225.27	1243.92
	12	1252.64	1254.38	1223.97	1235.71
	13	1233.81	1223.64	1209.02	1203.19
	14	1198.68	1161.12	1181.26	1147.83
	15	1147.02	1085.85	1144.99	1083.93
	16	1120.35	1053.19	1119.62	1050.47
	17	1119.47	1119.47	1118.73	1118.73
	18	1119.47	1119.47	1118.73	1118.73
	19	1119.47	1119.47	1118.73	1118.73
	20	1119.47	1119.47	1118.73	1118.73
	21	1119.47	1119.47	1118.73	1118.73
	22	1119.47	1119.47	1118.73	1118.73
	23	1119.47	1119.47	1118.73	1118.73

It was interesting to see a zone of green where the optimised trajectory improved performance, form around the yellow zone. However, this was less true for the cases when the sun was starting to set. This was because just like the previous simulations where the extra distance covered led to greater power being consumed than gained but because the sun is setting, the time taken to cover the extra distance is also costing the available capture of solar power over time. Thus, when the sun is setting, under the wrong conditions, the UAV's performance is punished greater for the optimised trajectory. Thus, under these conditions going in a straight line would be optimal.

While the reverse would be true for when the sun is rising as more time passes the greater power the UAV will capture. The question then becomes would the extra time spent on avoiding clouds to capture greater sunlight when going in a straight path beyond the 65km simulation plane be more beneficial to the overall UAV performance in the long run if the sun is already rising.

4.4 Optimal Strategy

With the aid of *table 6-8* it was easier to see the general optimum strategy to deploy. All that remained was to run more simulations to include every whole number in between 10 m/s to the maximum optimum speed. By utilizing the table *6-8* it was easier to find the minimum drag speed which was around 12m/s. One last dilemma remained and that was to decide whether under perpetual flight conditions (yellow regions) the strategy should aim for greater speed with the aid of modified trajectories. In the end it was concluded that for every extra m/s, the range of the UAV would be extended by 3.6 km per hour, thus, if the modified trajectory required deviation distance greater than 3.6km it was deemed unnecessary. Thus, in most perpetual flight simulations, trajectory remained unmodified, and the UAV stuck to a straight path.

Table 7: Optimal flight strategy during summer solstice

Hour	Speed	Final Battery Level	Power Gained	Battery Lost
0	12	1130.11	0	-144.89
1	12	1130.11	0	-144.89
2	12	1130.11	0	-144.89
3	12	1131.7	0	-143.3
4	12	1226.72	49.61	-48.28
5	15	1271.19	0	-3.81
6	15	1275	0	0
7	17	1275	0	0
8	20	1275	0	0
9	20	1275	0	0
10	21	1275	0	0
11	21	1275	0	0
12	21	1275	0	0
13	21	1275	0	0
14	20	1275	0	0
15	20	1275	0	0
16	17	1275	0	0
17	15	1275	0	0
18	12	1222.62	0	-52.38
19	12	1153.39	0	-121.61
20	12	1130.11	0	-144.89
21	12	1130.11	0	-144.89
22	12	1130.11	0	-144.89
23	12	1130.11	0	-144.89
Total Distance:	1350 km	313.2 km extra	49.61	-1383.61

Table 9 gives the final general optimal flight strategy during a summer solstice. The strategy is a general recommendation considering all the results from the 5 trajectories tested, and the recommended flight speed per hour is given. During summer solstice only in between 4am to 5am is a modified trajectory recommended helping save 49.61 Wh of energy. Now, how much time and other factors that had to be sacrificed to fly the modified trajectory to save the amount of battery stated remains to be further investigated. But under the recommended flight strategy 313.2 km of extra range (a total of 1350km in a day) is covered than when flying at the minimum drag speed of 12 m/s (1036.8 km). Lastly, in periods between the perpetual flight conditions (yellow zones) a total of 1383.61 Wh is expected to be lost which is the summation of all the power lost for each simulation. Given speed of 18 m/s or less takes more than an hour to fly 65km, 1383.61 Wh wouldn't be an accurate value of the total power consumed in between the yellow conditions, thus, perpetual flight would still be possible, as proven by the previous project on Limosaero's UAV [11].

Table 8: Optimal flight strategy during equinox

Hour	Speed	Final Battery Level	Power Gained	Battery Lost
0	12	1130.11	0	-144.89
1	12	1130.11	0	-144.89
2	12	1130.11	0	-144.89
3	12	1130.11	0	-144.89
4	12	1130.11	0	-144.89
5	12	1132.48	0	-142.52
6	12	1211.88	37.25	-63.12
7	13	1275	35.76	0
8	14	1275	0	0
9	15	1275	0	0
10	17	1275	0	0
11	18	1275	0	0
12	18	1275	0	0
13	17	1275	0	0
14	16	1275	0	0
15	13	1272.37	0	-2.63
16	12	1231.05	0	-43.95
17	12	1167.42	0	-107.58
18	12	1130.47	0	-144.53
19	12	1130.11	0	-144.89
20	12	1130.11	0	-144.89
21	12	1130.11	0	-144.89
22	12	1130.11	0	-144.89
23	12	1130.11	0	-144.89
Total Distance:	1155.6 km	118.8 km extra	73.01	-1953.23

Next is the optimal flight strategy during the equinox period. With the modified strategy the range can be extended by 118.8km, a total of 1155.6km in a day. It is recommended to apply modification to the trajectory to avoid clouds in between 6am to 8pm which would result in roughly 73.01 Wh of energy gained. Achieving perpetual flight seems to be harder during this period of the year as seen in the increased estimate of decrease in battery level (1953.23Wh) as when compared to during the summer solstice.

Table 9: Optimal flight strategy during winter solstice

Hour	Speed	Final Battery Level	Power Gained	Battery Lost
0	12	1130.11	0	-144.89
1	12	1130.11	0	-144.89
2	12	1130.11	0	-144.89
3	12	1130.11	0	-144.89
4	12	1130.11	0	-144.89
5	12	1130.11	0	-144.89
6	12	1130.11	0	-144.89
7	12	1137.67	0	-137.33
8	12	1168.72	0	-106.28
9	12	1239.75	36.02	-35.25
10	12	1257.65	28.91	-17.35
11	12	1254.44	14.77	-20.56
12	12	1241.93	6.71	-33.07
13	12	1221.05	2.28	-53.95
14	12	1188.97	0	-86.03
15	12	1151.95	0	-123.05
16	12	1131.03	0	-143.97
17	12	1130.11	0	-144.89
18	12	1130.11	0	-144.89
19	12	1130.11	0	-144.89
20	12	1130.11	0	-144.89
21	12	1130.11	0	-144.89
22	12	1130.11	0	-144.89
23	12	1130.11	0	-144.89
Total Distance:	1036.8 km	0 km extra	88.69	-2785.3

For winter solstice where perpetual flight conditions (yellow zones) are non-existent, it is recommended in between 9am to 2pm to modify its trajectory for cloud avoidance and most importantly, to always fly at minimum drag speed. Because of this strategy the range of the UAV isn't improved significantly, however, it is expected that the strategy would lead to roughly 88.69 Wh of energy to be saved which could be utilized to extend the flight duration thus the range.

Simulations during winter solstice demonstrates a greater impact of cloud cover when solar irradiance is scarce thus, to get a step closer to achieving perpetual flight all season long, greater understanding and research into cloud avoiding trajectory could be crucial.

Discussion

5.1 Interpretation of the Result

The results section effectively achieved the aim of identifying the optimal flight strategy during different celestial events—summer solstice, equinox, and winter solstice—by manipulating the UAV's speed and trajectory. It was determined that during periods of 0 solar irradiance, flying at the minimum drag speed of 12 m/s was optimal. However, during the summer solstice, speeds could be increased to 21 m/s, 18 m/s during the equinox, and maintained at 12 m/s for the winter solstice around noon times of peak solar irradiance.

Regarding trajectory variation, it was found that maintaining a straight line was generally the most efficient course of action, except during specific time intervals: 4am to 5am during the summer solstice, 6am to 8am during the equinox, and 9am to 2pm during the winter solstice. These were under the assumption that deviating from the straight path during these periods allowed for faster travel time given the extra distance going off course didn’t sacrifice range.

5.2 Limitations of the Result

It is important to note that all results were obtained from individual simulations, meaning that each data point / final battery level represented a separate instance. For instance, when flying at 10 m/s, the simulation required more than 1 hour to cover 65km therefore the next simulation starting 1 hour later isn’t a continuous simulation event but an exclusive event thus, the result is discontinuous. Consequently, technically the simulation failed to identify the best continuous optimal flight strategy. The primary limitation of the results lies in the uncertainty surrounding the optimal flight strategy. While the suggested strategies may conserve battery life and extend flight range, they also entail sacrificing time to achieve these gains. Achieving a definitive optimal flight strategy requires further investigation and refinement.

24-hour simulation

To truly evaluate whether the battery gains outweigh the time losses, the simulation should encompass an entire day, necessitating a larger number of cloud images or a single large cloud image extensive enough for the UAV to traverse over 24 hours. However, using a stationary large cloud image poses challenges, as clouds are rarely static and often move at speeds comparable to or faster than the UAV. Clouds patterns would also change over time.

Dimension measurement

Furthermore, the dimensions that each cloud represented was measured in a way that left many uncertainties using subjective measures to determine the dimensions. Additionally, relying on just five cloud images based on their desirable qualities that fit the needs of the project may have introduced biases or inaccuracies into the analysis, particularly in scenarios where cloud characteristics vary spatially or temporally.

Cloud cover uncertainty

One significant simplification in the project was the assumption that the densest part of the cloud would block out 70% of sunlight. This assumption carries risks for several reasons. Firstly, it was assumed that a specific RGB value (r=142, g=142, b=142) represented the lightest pixel, but the sample image data only spanned a week, potentially missing images with lighter clouds during stormier seasons. Secondly, assuming a linear relationship between pixel colours throughout the cloud's intensity range is unsupported, and it's possible that pixel values may follow other patterns in between the coefficient values 0.30 and 1.

Furthermore, the two-dimensional image used in the analysis lacks information about cloud height variation. These variations could significantly impact solar irradiance, underscoring the need for further research and investigation into this aspect.

Seconds over minutes

The choice of minutes over seconds for simulation updates, while beneficial for a lighter and smoother simulation, negatively impacted UAV trajectory accuracy. Each simulation update representing a minute led to common issues such as overshooting coordinates due to the large variation in the fixed distance that the UAV moved each minute for the various speeds.

Addressing these limitations and conducting further research into the complex dynamics of UAV flight under varying environmental conditions are crucial steps towards developing more efficient and reliable flight strategies for solar-powered UAVs.

5.3 Further investigation & improvement

The project has immense potential for further development and improvement by incorporating many more realistic conditions for more accurate and relevant results.

Incorporating longitude

The equations used to calculate daily solar irradiance were simplified in the project, with the longitude component disregarded for simplicity. However, incorporating longitude as an input could enhance irradiance accuracy, providing more precise sunrise and sunset times. Additionally, updating longitude during simulation runtime based on the UAV's location could be explored. Given the distance covered at minimum drag speed is 1036.8km, which is almost the distance from Bath to Berlin, the UAV is likely to experience some sort of time shift/ jet lag during flight if flying west or east which would affect the solar irradiance over time.

Explore results and various latitudes and other factors

Achieving perpetual flight conditions at higher latitudes in the UK posed challenges due to reduced solar power availability on most days. With additional time, implementing a more automated code to explore various parameters such as latitudes, altitude, and speed could have yielded a richer dataset for analysis. Incorporating variable speed would be particularly

beneficial, allowing for a more nuanced understanding of how acceleration and deceleration impact performance. This approach could help bridge the gap between the discrete speed values per hour suggested in the project and provide valuable insights into optimizing flight dynamics under changing environmental conditions.

Optimal Flight Trajectory

In the project, trajectories were modified in a simplistic manner, permitting only one deviation move. These modifications were based on subjective assessment, lacking accuracy, precision, and a consistent rationale for deviation from the main route. This area is subject for further exploration to minimize off-course distance while maximizing cloud avoidance benefits. Determining which cloud intensity coefficients should be avoided and which are flexible based on location is crucial for optimizing flight paths.

Addition of wind

Next step for further research would be the implementation of wind on top of clouds. This addition would pose another great challenge and a realistic problem the UAV may encounter.

Shifting clouds

The primary limitation of the project was the confined 65km plane, restricting the simulation from exploring beyond this range. Since the project aimed to analyse optimal flight strategies over an entire day, ideally, future simulations should enable continuous UAV operation for 24 hours. To achieve this, the simulation would require multiple cloud images to render, eliminating the constraints of a single image. Finally, the simulation should be able to incorporate the effects of moving clouds by shifting the grid of the cloud intensity coefficient by the speed and direction in which the cloud moves. This will give a new avenue to explore how the UAV's direction of movement relative to the cloud's motion affect the ideal trajectory of the UAV as moving against the clouds would help the UAV bypass them much faster than moving in the same relative direction as the clouds.

Conclusions & Future Work

The research proposed to accomplish the following aims:

- Investigate factors that affect solar power available to the UAV such as cloud coverage, time of year, and location of the UAV.
- Develop a flight model in 2 degrees of freedom to explore how various speed & UAV trajectory modification affects the UAV's performance at various times of the year.
- Determine the optimal flight strategy over a day by varying the speed and trajectory of the UAV.

The project focused on assessing the optimal flight strategy for a solar-powered UAV across different seasonal conditions, speeds, and trajectories. Due to time constraints, the project concentrated on latitude 44.5 degrees and tested six speed configurations (5, 10, 15, 20, 25, 30 m/s). The simulations aimed to determine the most efficient flight strategy by evaluating the UAV's battery level upon completion of a 65km flight path.

Initially, the impact of solar irradiance and speed on battery levels was examined. Results indicated that flying at the minimum drag speed (12 m/s) was optimal under conditions of no sunlight. However, during periods of abundant sunlight, faster speeds could be sustained without depleting battery levels, particularly during the summer solstice where top speeds of 21 m/s could be reached, 18 m/s for equinox period and sustained 12 m/s during winter solstice.

These strategies extended the UAV's range by 313.2 km during summer solstice, 118.8 km during equinox, and maintained the same range during winter solstice.

Another set of simulations explored the effects of clouds on battery levels under constant solar irradiance conditions. It was observed that while adjusting trajectory to avoid clouds could improve performance under certain conditions, it could also lead to greater power consumption, especially when solar power availability was insufficient.

Further analysis involved testing the optimized trajectory under normal day cycle conditions. The simulations revealed instances where modified trajectories improved performance, particularly when solar power became ample during sunrise hours. However, during sunset, the extended flight time required for modified trajectories could outweigh the benefits of cloud avoidance as the extra flight time led to the sun setting more.

Based on the results, an optimal flight strategy was proposed for each season. During the summer solstice, it was recommended to modify trajectory between 4am to 5am to save energy, while maintaining a straight path during other hours. Similarly, trajectory adjustments were suggested during the equinox in between 6am to 8am for equinox and 9am to 2pm for winter solstice, with an emphasis on flying at minimum drag speed for the entire duration during the latter.

By modifying trajectory, power savings of 49.61 Wh, 73.01 Wh, and 88.69 Wh could be achieved during summer solstice, equinox, and winter solstice, respectively.

The implications of these findings are profound for the future development of UAV technologies, especially for applications that demand high endurance and reliability under varying environmental conditions, such as environmental monitoring, disaster response, and agricultural surveillance. The ability to extend operational time without the need for frequent recharging or refuelling can dramatically increase the UAV's utility in continuous data collection and real-time monitoring tasks.

Future research should build on these findings by incorporating more dynamic models of environmental change, particularly models that predict cloud movement and solar exposure more accurately. Testing these optimized flight strategies in real-world conditions will be crucial to validate the simulation results and to refine the strategies for practical application. Moreover, integrating machine learning algorithms could enable UAVs to autonomously adjust their flight plans in response to instantaneous changes in weather conditions, further enhancing their efficiency and application range.

Areas for further research were identified in the discussion session:

- Utilization of advanced studies into solar irradiance and effects of cloud cover under more realistic conditions.
- Identification of ideal trajectories to balance the UAV's ability to capture additional solar irradiance capture while minimizing the penalties associated by going off course.
- Effects of moving clouds and how it would affect the UAV's performance and trajectory modification.
- Development for a largescale 24-hour simulation with accurate cloud cover image without boundaries
- The consequences of extreme wind on the power consumption of the UAV

In conclusion, this study not only advances our understanding of the operational dynamics of solar-powered UAVs but also sets the stage for the development of more sophisticated, autonomous, and environmentally adaptive UAV systems. By continuing to explore and address the challenges identified through this research, the potential for UAV technology to support a wide range of industrial, environmental, and humanitarian efforts can be fully realized, leading to smarter, more sustainable aerial solutions.

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Appendices

Link to the Excel file containing all the results of the simulations:

https://d.docs.live.net/fdd192bf161fb1f6/Desktop/FYP_results_khl71.xlsx

Link to GitHub containing all the relevant Python codes involved in the project:

https://github.com/kihoon2000/Limosaero.git